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# Determination of total extractable organofluorine (EOF) in food contact materials and target and non-target analysis of per- and polyfluoroalkyl substances using LC–MS/MS and LC–HRMS simultaneously coupled to ICP-MS

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**ABSTRACT**

Food contact materials (FCMs) from three countries were analysed for all extractable organofluorines (EOFs) from the materials and subsequently by target and non-target analysis for per- and polyfluoroalkyl substances (PFAS). The EOF varied by two orders of magnitude for FCM from UK and Saudi Arabia ranging between 2.14 and 483 ng cm$^{-2}$ (0.2–48 ng g$^{-1}$) showing that one quarter of all samples were above the Danish regulation for PFAS in FCM. Target PFAS showed high variability in composition and accounted for less than 1% of the EOF. Non-target PFAS screening using HPLC–ICP-MS and coupled simultaneously to HRMS showed the occurrence of organofluorines which were identified by neither LC–MS/MS nor LC–HRMS. This illustrates that the current target PFAS approaches fail to identify EOFs from FCM, which would be a problem with the new EU proposal to ban all PFAS.

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## Introduction

Per- and polyfluoroalkyl substances (PFAS) are a group of human-made organic chemicals in which all or most hydrogen atoms are replaced by fluorine (F) (Buck et al. 2011) or, according to the new OECD definition, have at least one fully fluorinated methyl or methylene carbon (OECD 2021). They have both hydrophobic and lipophobic properties. Because of the C–F bond, the fluorocarbon chain is extremely resistant to heat and is chemically stable (O'Hagan 2008). During recent decades PFAS have been used in various industrial products as well as in consumer products including paper raw materials (Jovanović et al. 2024) often used for food contact materials (FCMs) such as paper plates, beverage cups, wrapping paper, milk cartons, microwave popcorn bags and pizza boxes (Zafeiraki et al. 2014) to improve moisture and oil resistance (Schultes et al. 2019; Curtzwiler et al. 2021).

PFAS accumulate in a wide range of human tissues, including blood serum, umbilical cord blood, liver tissue, seminal fluid and breast milk (Eriksen et al. 2011; Sun et al. 2017). They pose great threats to human health and research on environmental pollution and health risk of PFAS has become a hot topic in recent years (Butt et al. 2010). PFAS accumulated in humans exhibit various toxicities such as hepatotoxicity, embryotoxicity, reproductive toxicity, neurotoxicity, endocrine toxicity and carcinogenicity (Lau et al. 2007; Sun et al. 2017). On entering the body, PFAS are

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poorly eliminated. The estimated serum half-lives of perfluorohexane sulphonic acid (PFHxS), perfluorooctane sulphonic acid (PFOS) and perfluorooctanoic acid (PFOA) in humans are 5.3, 4–5 and 3.5 years, respectively (Olsen et al. 2007; Ullah et al. 2014; Li et al. 2018).

The European Food Safety Authority (EFSA) released a recommendation for the limit of combined exposure to four PFAS [PFOA, perfluorononanoic acid (PFNA), PFOS and PFHxS] in food to 4.4 ng $kg^{-1}$ bodyweight per week (EFSA 2020; EU 2020). The risk of bioaccumulation has led to the listing of PFOS, perfluorooctane sulphonyl fluoride (PFOS-F) and recently PFOA as persistent organic compounds in the Stockholm Convention 2009 Annex B (Wang et al. 2009). However, their homologues, neutral precursors and emerging PFAS such as the perfluoroalkyl phosphonates (PFPA), perfluoro sulphonamides and fluoropolymers continue to be produced. Hence, the EU recently proposed the ban of all PFAS (ECHA 2023).

Even though PFAS polymers are said to pose less risk to human health, their production contributes to the presence of non-polymers in the environment (Blake et al. 2020). In the European Union, regulations regarding PFAS in FCM fall under the framework of Regulation (EC) (No. 1935/2004 2004), requiring that FCM do not transfer their constituents to food in quantities that could endanger human health, change the composition of food, or deteriorate its organoleptic characteristics. PFAS were detected in FCM, packaging and textiles with the most abundant ones being C4–C12 perfluoro-carboxylic and sulphonic acids (PFCA and PFSA), PFPA, fluorotelomers (FTOH) and perfluoro sulphonamides. Zafeiraki et al. (2014) detected perfluorotridecanoic acid (PFTrDA), perfluorotetradecanoic acid (PFTeDA) and trifluorohexadecanoic acid (PFHxDA) in fast food boxes from a Greek market, while perfluorohexanoic acid (PFHxA) was found in ice cream cups and several PFAS were detected in fast food wrappers. PFOA and PFOS were observed in microwave popcorn bags at the ng $g^{-1}$ level (Begley et al. 2005; Sinclair et al. 2007; Dolman and Pelzing 2011; Martínez-Moral and Tena 2012; Poothong et al. 2012) and 33.7–81.3 µg $kg^{-1}$ in textiles (Gang et al. 2009).

Despite the widespread use of PFAS in FCM, only a few studies have been published on concentrations of PFAS in FCM with non-target analytical methods, even though the identified fraction of PFAS in the previous studies accounted only for up to 5.5% of the extractable organofluorine (EOF) (Schaider et al. 2017). In this study we made use of the method developed by Jamari et al. (2017) where fluorine was detected as a polyatomic $BaF^+$ at $m/z$ 157 to do non-target fluorine analysis of PFAS. The separation of different classes of PFAS was done by HPLC which is simultaneously coupled to the ICP-MS/MS and high resolution ESI-MS for non-target analysis as demonstrated recently (Heuckeroth et al. 2021). Moreover, in order to identify whether target analysis of PFAS truly reflects the exposure of PFAS to humans from FCM, and for fluorine mass balance purposes we determined the EOF of food packaging from different countries [UK and Kingdom of Saudi Arabia (KSA)] using high resolution graphite furnace molecular absorption spectrometry (HR-GFMAS) (Akhdhar et al. 2020). Lastly, we performed quantification of target species using LC–MS/MS for the most common PFAS to perform a mass balance for EOF compounds FCM. The aim is to determine target PFAS in the EOF and demonstrate how much PFAS is missed when only target analysis is performed.

## Material and methods

### *Sample collection*

To check the variability of FCM, 36 samples from three different countries (Sweden, KSA and UK) were collected from the market. Since composition might differ by outlet and usage, a variety of FCM were chosen. Sample description can be found in Table S1 in the supporting information (SI).

### *Extraction of FCM samples*

FCM samples were each cut into 5 × 5 cm pieces, then placed into glass containers. 40 mL of methanol and polypropylene stirrer bars were added to each container. Samples were stirred for 8 h at

room temperature, then the extract was transferred to smaller glass vials and evaporated under nitrogen to dryness and then 500 µL of MeOH were added. The extracted samples were then transferred to 2 mL Eppendorf tubes with 25 mg ENVI-Carb (Merck, Germany), vortex-mixed and centrifuged at 10,000 rpm for 10 min. An aliquot of the extract was transferred to HPLC vials and taken for analysis.

### Instrumental analysis

#### EOF analysis by HR-GFMAS

EOF was analysed according to the method described by Akhdhar et al. (2020). An Analytik Jena ContrAA 700 High Resolution Continuum Source Atomic Absorption Spectrometer with a transversely heated graphite tube atomizer was used for all measurements (details in 'Analytical methods' section in SI and Table S2). The 606.417 nm wavelength was used to detect fluorine as CaF.

Only samples from KSA and UK were measured for total fluorine (TF) in the extract as EOF and reported as mass fraction of fluorine per surface area. Not enough material was available for the Swedish samples to perform TF in EOF.

#### Non-target PFAS screening for identification

Samples were analysed by HPLC–ICP-MS/MS–ESI-HRMS according to the method described by Jamari et al. (2019) (details in SI 'Analytical methods' section). The separation was performed on an Agilent 1290 LC system. After separation, analytes were simultaneously detected using an 8800 Triple Quadrupole ICP-MS/MS (Agilent Technologies, UK) and an ESI-HRMS (LTQ Orbitrap Discovery, Thermo Scientific), operated in negative mode. The eluent was split with 80% to the ICP-MS/MS and 20% of the flow to the ESI-HRMS.

For ICP-MS/MS, $F^-$ was detected as $BaF^+$ ($m/z$ 157) and to enable formation of $BaF^+$, $F^-$ containing standards or samples were mixed post-column with 50 mg $L^{-1}$ $Ba^{2+}$ online via a T-piece at a mixing ratio of 1:1.5. The instrument was tuned to maximum signal to background ratio using oxygen as a reaction gas at $m/z$ 157 ($^{138}Ba^{19}F^+$). Sulphur was also monitored using $m/z$ 48 ($^{32}S^{16}O^+$). Parameters for ICP-MS/MS–ESI-HRMS are shown in Table S3. The ESI-HRMS was used in a non-target analysis mode screening from $m/z$ 50 to $m/z$ 1000. The most intense peaks were also fragmented and MS2 spectra were recorded. The identification of compounds was by accurate mass (M − H) and the molecular formulae was accepted when it showed better than ±5 ppm error.

#### Target PFAS analysis for quantification using LC–MS/MS

For target analysis quantification was done using a triple quadrupole mass spectrometer (QSight 220 mass spectrometer, PerkinElmer, UK) operated in negative electrospray ionisation mode for 19 different PFAS. The instrument was optimised for multi-reaction monitoring transitions of the different PFAS. The chromatographic method used was described previously in Nxumalo et al. (2023). A detailed description can be found in the SI (Figure S1 and Table S4).

### Quality control

For target analysis, calibration curves were prepared in methanol ranging between 0.05 and 50 µg F $L^{-1}$ using 2 mg $L^{-1}$ PFAS MXC mix (Wellington Labs, Guelph, ON, Canada). The limit of detection (LOD) and limit of quantification (LOQ) (µg F $L^{-1}$) were calculated from the calibration curves for the individual PFAS as 3.3× and 10× the standard error of the $y$ intercept divided by the slope. Procedural blanks (negligible, <5% of FCM results) were also processed and subtracted from the samples. Seven samples were extracted as duplicates to assess the precision using a calculated relative standard deviation given in Table S5.

For EOF analysis, calibration curves ranging from 0.2 to 4 mg $L^{-1}$ were prepared in methanol from PFOA standard solution. LOD and LOQ were calculated as described above.

Recoveries were calculated using plain printing paper ($n = 3$) spiked with 1 mL mixture containing six PFAS [PFHxA, PFOA, perfluorodecanoic acid (PFDA), PFHxS, PFOS and perfluorooctane sulphonamide (PFOSA)] in methanol. These samples

were then extracted as described above including evaporation and reconstitution, then analysed using by the three different analytical methods: HPLC–ESI-HRMS, HR-GFMAS and HPLC–ICP-MS/MS. The spike concentration was chosen so that 100% recovery would result in a concentration of 87 µg F $L^{-1}$ for each PFAS after the addition of 40 mL MeOH extractant. The spike is equivalent to 139 ng $cm^{-2}$ or 13.9 ng $g^{-1}$. The data in this study were calculated per surface area, but in places mass fractions are given and here 100 g FCM $m^{-2}$ is assumed.

## Results and discussion

### Quality control

Recoveries for the different compounds from the spiked samples determined using external calibration, HPLC–ICP-MS/MS and the HPLC–ESI-HRMS were between 61.1% and 133%, (Figures S2 and S3 and Table S6) with no significant difference between ICP-MS/MS and ESI-HRMS (*t*-test, $p = 0.4$). The sum of the different PFAS spiked on plain paper gave a recovery of 93% for the HPLC–ICP-MS/MS and 90% for HPLC–ESI-HRMS, while the recovery using HR-GFMAS was slightly higher (130%) (Figure S3). The reason for the higher recovery is unknown but HR-GFMAS is able to detect compounds which cannot elute from an HPLC column (e.g. F-polymer or F-containing particles). Additionally, higher F-concentrations measured by HR-GFMAS were also observed in comparison with combustion ion chromatography (CIC) when Gehrenkemper et al. (2021) analysed surface water and Müller et al. (2023) analysed ski wax. Since no CRM is available for PFAS in paper, more investigations are obligatory to establish the correct value. HR-GFMAS was further used for the determination of TF in the EOF fraction in this study. Duplicate extraction of FCM samples showed compound-dependent error of between 19% and 141% (see Table S5). The extraction had previously been optimised (i.e. further extractions did not lead to higher EOF). The variability might be a reflection on the heterogeneity of the samples, since the precision of the paper samples spiked with PFAS was much better (RSD < 10%) for most compounds. Therefore, either taking larger pieces of FCM, which makes the extraction with more solvent cumbersome, or fluorine mapping of the surface, such as particle-induced gamma emission (PIGE) (Schultes et al. 2019) or laser ablation ICP-MS (Clases et al. 2023) would be appropriate analytical techniques to obtain a good insight how heterogeneously fluorine is distributed on the paper and could inform how paper should be sampled to get a representative analysis of the FCM.

### Target analysis using LC–MS/MS

A total of 16 target PFAS were detected in the different FCM samples. The majority of the detected PFAS were PFCA with carbon number ≤8. This is consistent with the observations from the non-target ESI-HRMS, which could only detect PFAS with a carbon chain length of C8 or shorter (described later). The sum of the individual PFAS ranged from 0.02 to 37.1 ng F $cm^{-2}$ (equivalent to 0.002–3.7 ng $g^{-1}$) as seen in Figure 1. It shows the distribution of the individual PFAS (PFOSA is not included in the plot since it gave very high concentrations from FP 8 (2.2 ng F $cm^{-2}$) and FP 9 (37.0 ng F $cm^{-2}$) (Table S8). PFBA and PFHxA were the most common PFAS and were present in all samples. It should be noted that, even though those compounds were found in all samples, their concentrations were lower than the concentrations of PFOS and PFOSA which were in some samples 10 times higher. Generally, samples from the KSA had higher concentrations of PFOS and PFHxS than the UK samples. The KSA samples also displayed generally a higher sum of the identified PFAS, so more target PFAS (Table 1). The high concentrations of PFOS in the KSA samples reflected probable regional difference in production patterns compared to similar samples taken from the UK (Table 1).

Target analysis using LC–MS/MS using spiked isotopically labelled PFAS standards was performed on all the EOF samples. The target analytes included perfluoro carboxylates (PFCA; C4–C15), perfluoro sulphonates (PFSA; C4–C10), N-ethyl perfluorooctyl sulphonamide (N-EtFOSA), PFOSA, fluorotelomer carboxylic acids (FTC; 5:3, 7:3, 6:2, 8:2 and 10:2), fluorotelomer sulphonic acids (FTS; 2:2, 4:2, 6:2, 8:2, 10:2), perfluoroalkyl

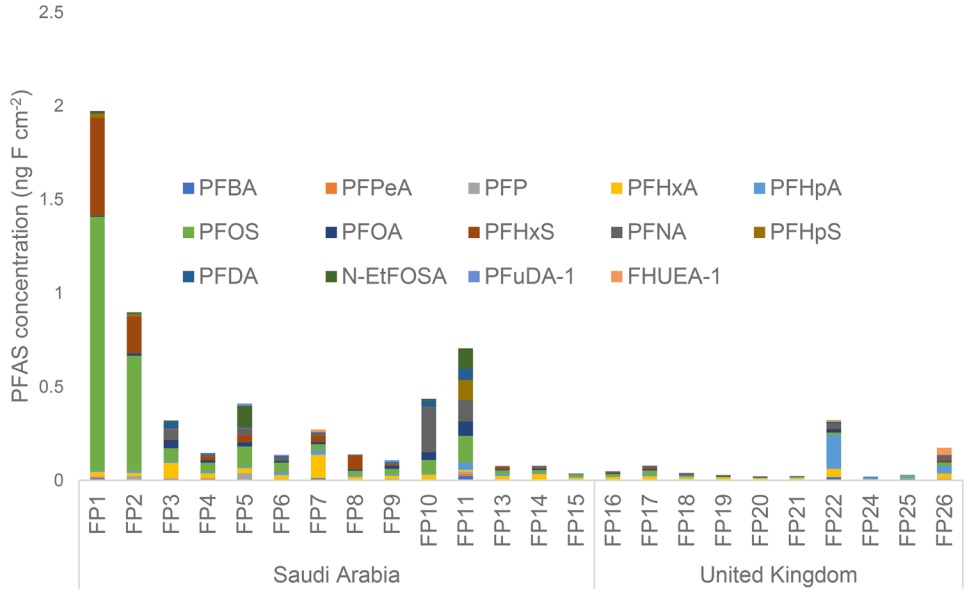



**Figure 1.** Identified PFAS plotted as ∑PFAS (measured using target LC–MS/MS) for KSA and UK samples in which individual PFAS were quantified. Excluded are additional PFOSA > 2 ng cm$^{-3}$ for FP 8 and FP 9.

**Table 1.** EOF mass balance in FCM samples using HR-GFMAS for EOF and LC–MS/MS for target analysis.

| Country | Sample ID | Sum of identified PFAS (ng F cm$^{-2}$) | EOF concentration (ng F cm$^{-2}$) | $C_{F_\sum PFAS}$/EOF (%) |
|---|---|---|---|---|
| Kingdom of Saudi Arabia | FP 1 | 1.99 | 288 | 0.7 |
| | FP 2 | 0.98 | 99.3 | 1.0 |
| | FP 3 | 1.08 | 17.7 | 6.1 |
| | FP 4 | 0.40 | 44.5 | 0.9 |
| | FP 5 | 0.41 | 10.6 | 3.9 |
| | FP 6 | 0.43 | 60.0 | 0.7 |
| | FP 7 | 1.03 | 240 | 0.4 |
| | FP 8 | 2.36 | 267 | 0.9 |
| | FP 9 | 37.1 | – | – |
| | FP 10 | 0.72 | 89.8 | 0.8 |
| | FP 11 | 0.73 | 34.1 | 2.2 |
| | FP 13 | 0.24 | 34.9 | 0.7 |
| | FP 14 | 0.47 | 273 | 0.2 |
| | FP 15 | 0.10 | 32.5 | 0.3 |
| United Kingdom | FP 16 | 0.20 | 41.3 | 0.5 |
| | FP 17 | 0.21 | 7.64 | 2.8 |
| | FP 18 | 0.04 | 483 | 0.0 |
| | FP 19 | 0.15 | 36.4 | 0.4 |
| | FP 20 | 0.02 | 32.5 | 0.1 |
| | FP 21 | 0.14 | 162 | 0.1 |
| | FP 22 | 0.48 | 56.8 | 0.9 |
| | FP 24 | 0.18 | 34.5 | 0.5 |
| | FP 25 | 0.13 | 205 | 0.1 |
| | FP 26 | 0.25 | 24.7 | 1.0 |

phosphonic acids (PFPA; C4, C8), perfluorinated ether sulphonates (PFAES; 4:2, 6:2, 8:2, 10:2) and chlorinated perfluorinated ether sulphonates (Cl-PFAES 4:2, 6:2, 8:2). The neutral loss of 44 to give the fragment $[M–CO_2–H]^-$ can be used to identify the presence of carboxylic acids (Berger et al. 2004; Zabaleta et al. 2015). This fragment was used as confirmation for PFCA. PFOS, PFOSA and perfluorooctylphosphonic acid (PFOPA), which were also confirmed using analytical standards. FTC, FTS, PFAES, Cl-PFAES were not found in any of the samples.

Samples from Sweden had the greatest number of different classes of PFAS. In addition to the PFCA and PFSA which were found in the KSA and UK samples, N-EtFOSA, PFOSA and PFPA were also determined in some of the Swedish samples. Only one of the samples, FP 32, which is a microwave popcorn bag, had none of the target compounds. PFOS, PFHxA and PFOA were very common in the Swedish samples, since they occurred in more than 70% of the total samples, which was not the case for the UK and KSA samples.

PFCA (C4, C6–C10), PFOS and PFOSA were identified in one of the Swedish samples (McDonald's French fries cardboard box), while no PFAS was identified from a similar sample bought in KSA. This might indicate differences in the supply chain for those packaging materials. PFHxA was the only compound found in a pizza box from the UK. PFOA, PFHxS, perfluoroheptanoic acid (PFHpA), perfluoroheptanesulphonic acid (PFHpS), PFOS, PFOSA, N-EtFOSA were identified in most of the microwave popcorn bags from Sweden (Figure 2), while only PFOA was determined in the KSA microwave popcorn bag. PFOA in popcorn bags were determined by Begley et al. (2005) and the concentrations ranged

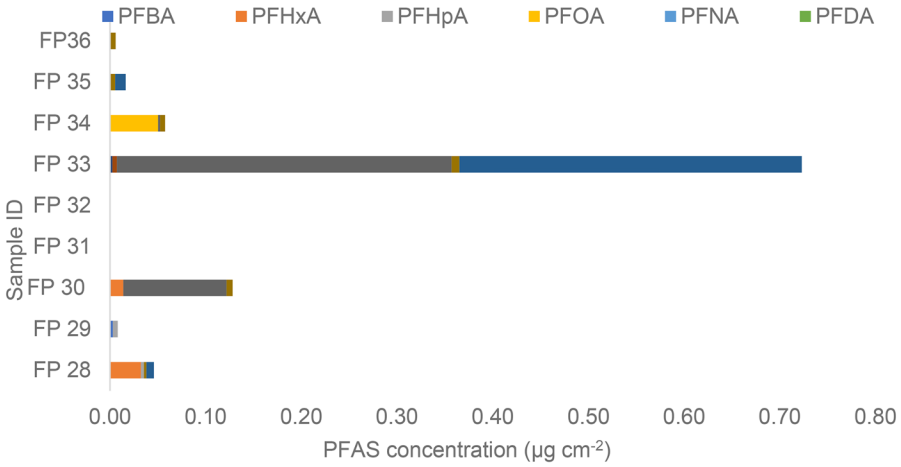



**Figure 2.** Concentrations of the different PFAS compounds in FCM samples purchased in Sweden. Quantification was done using HPLC–ESI-HRMS data.

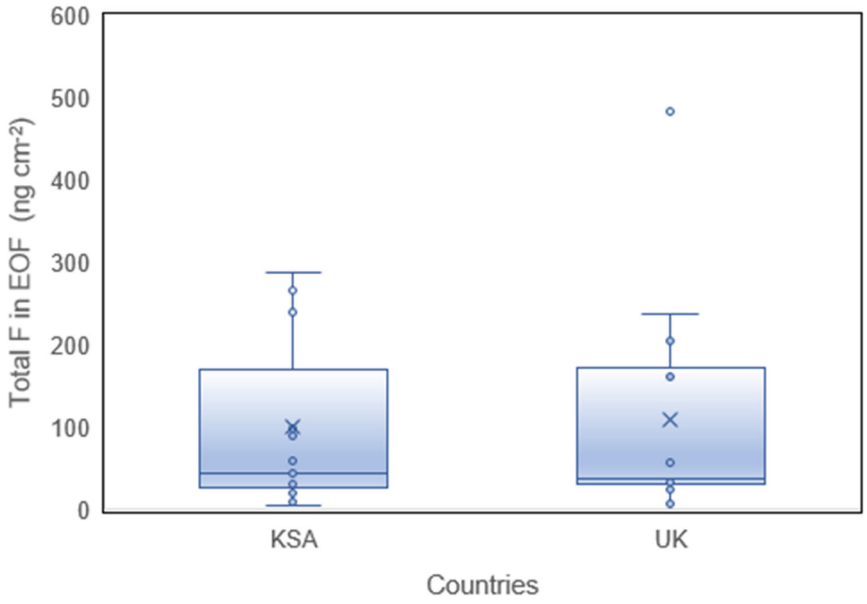



**Figure 3.** EOF concentration on food packaging in KSA and UK samples using HR-GFMAS.

from 6 to 290 ng g$^{-1}$ while in this study the concentrations were between 148 and 32,270 ng g$^{-1}$. Target PFAS were not identified in 15 (41.6%) of analysed food packaging samples.

### *EOF*

All FCM showed measurable EOF content. Concentrations of EOF found in the KSA samples ranged from 4.40 to 289 ng cm$^{-2}$, while the levels observed in samples from UK were between 2.14 and 483 ng cm$^{-2}$. Hence, the concentrations ranged over two orders of magnitude in both countries (Figure 3). The highest EOF was found in a dessert wrapper (UK) at 483 ng cm$^{-2}$ equivalent to 48 ng g$^{-1}$). Other high concentrations from the KSA and UK samples were found in French fry wrappers (288 and 237 ng cm$^{-2}$ respectively). A previous study found EOF concentrations in a similar range (40–490 ng cm$^{-2}$) (Schultes et al. 2019).

Robel et al. (2017) and Schultes et al. (2019) detected TF levels of up to 8170 and up to 17,800 ng cm$^{-2}$ in packaging and FCM, showing that potentially much more PFAS can be found in FCM; however, they are not fully extractable with methanol. Fluoropolymers, for example, have been found to have low solubility in non-fluorinated solvents such as methanol

(Rankin and Mabury 2014; Schultes et al. 2019), hence higher concentrations of TF in FCM would be expected in the samples measured in this study compared to their EOF.

The sum of target PFAS was often less than 1% of EOF. Overall, 7 of the 27 (26%) samples were above the Danish regulation (10 µg of organic fluorine per $dm^2$ paper, Ministry of Environment and Food of Denmark 2020) which equals 100 ng $cm^{-2}$ for total EOFs (Table 1).

### *Non-target analysis using LC–ICP-MS/HRMS*

Since the majority of the organofluorines in the EOF extracts could not be identified and quantified through a target LC–MS/MS method, we employed here the novel approach of using simultaneous ICP-MS/MS as a fluorine-specific detector for HPLC separation in addition to a non-target analysis using accurate mass (ESI-HRMS). The identification of organofluorines was achieved by using fluorine-specific ICP-MS simultaneously coupled to ESI-HRMS. The HRMS data were checked only at retention times at which a fluorine peak was detected by ICP-MS/MS (Figure 4). Samples which showed fluorine peak (*m/z* 157, $BaF^+$) on the ICP-MS/MS were FP 30, FP 33 and FP 34 which were a French fry paper bag and microwave popcorn bag from Sweden (Figures 4 and S10). Additional to fluorine also sulphur can be monitored by ICP-MS/MS

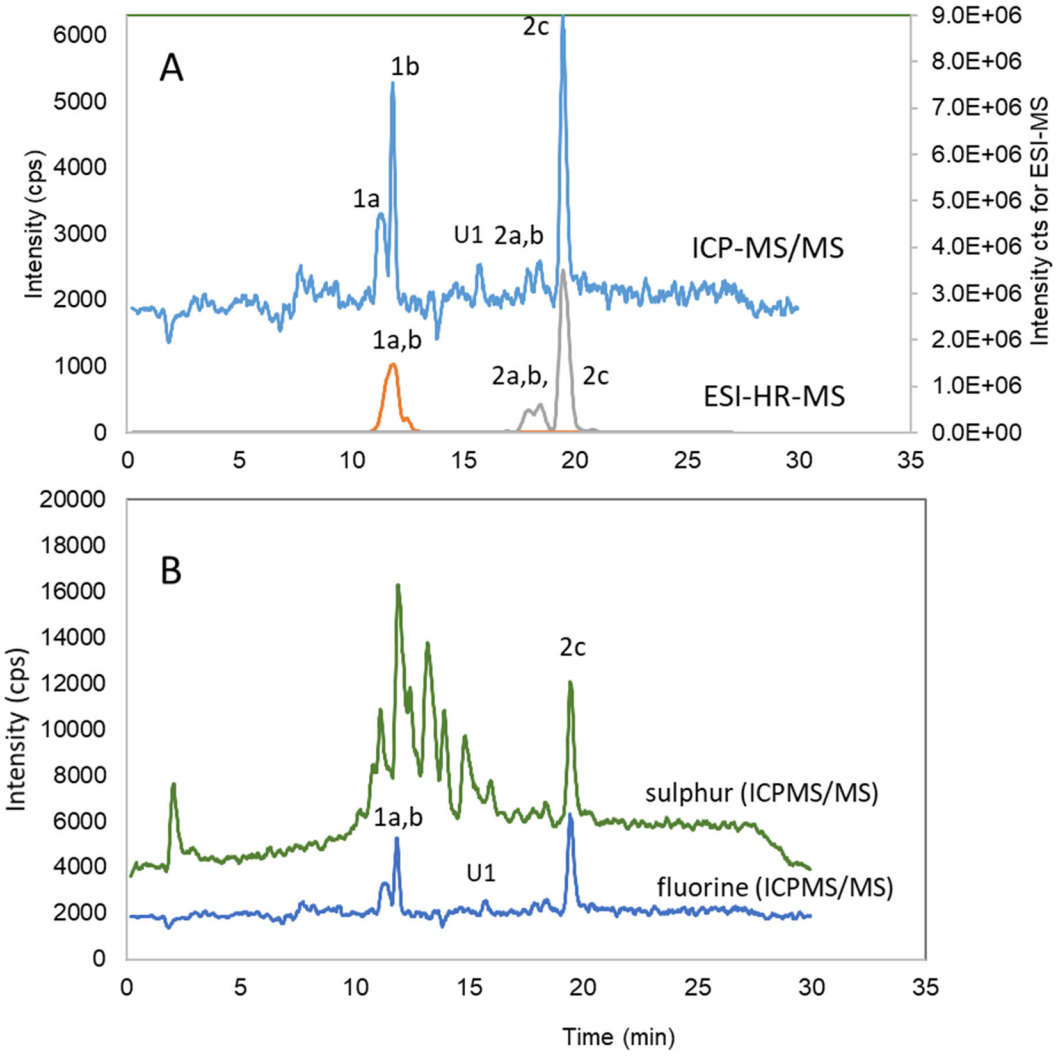



**Figure 4.** Chromatogram (A) of FP-33 using RP-HPLC and monitored online simultaneously for fluorine [blue; $BaF^+$ *m/z* 157 (ICP-MS/MS)] and other molecular masses by ESI-HRMS; here for branched and linear PFOS (orange, peaks 1a,b = *m/z* 498.92–498.94) and for branched and linear N-EtFOSA (grey, peaks 2a–c = *m/z* 525.96–525.98). (B) The sulphur detection (green; $SO^+$ *m/z* 48 of ICP-MS/MS) can be seen. U1 is an unidentified organofluorine compound.

(Figure 4(B)). For the ICP-MS/MS peaks which showed corresponding masses for possible PFAS, the extracted masses of these compounds are shown in Figure 4(A). The accurate masses recorded by the ESI-HRMS for peaks 1a,b and 2a–c are shown in Table 2. The masses and ion formulas of peaks 1a,b matched PFOS isomers (branched and linear), while peaks 2a–c matched that of N-EtFOSA isomers (Table 2). Two fragments commonly used to confirm the presence of sulphonic acids; $SO_3^-$ ($m/z$ 80.066) and $FO_3S$ ($m/z$ 98.955) (Lin et al. 2016; Zacs and Bartkevics 2016; Barzen-Hanson et al. 2017) were detected in peak 2c. Additionally, since both PFOS and N-EtFOSA contain sulphur (S), these compounds were further confirmed using the sulphur-specific detection of $SO^+$ ($m/z$=48) from the ICP-MS/MS which showed peaks at the corresponding $BaF^+$ retention times (Figure 4(B)). Ion mobility methods could clarify the isomers further (Dodds et al. 2020). Figure 4 showed also an unidentified organofluorine compound (U1), the corresponding ESI-HRMS data were scanned against a typical suspect PFAS list, but no compound could be identified.

The $m/z$ shown in Figure 4 were measured by ESI-HRMS; here for branched and linear PFOS (orange, peaks 1a,b = $m/z$ 498.92–498.94) and for branched and linear N-EtFOSA (grey, peaks 2a–c = $m/z$ 525.96–525.98). In Figure 4(B) the sulphur detection (green; $SO^+$ $m/z$ 48 of ICP-MS/MS can be seen. U1 is an unidentified organofluorine compound.

In another sample (FB 30) the fluorine peak did also not show any corresponding peak in the ESI-HRMS, for which only the ion count is shown in Figure S4. This could mean that the species associated with this peak does not ionise efficiently in negative and positive mode in an ESI source but was extractable with methanol. Despite the relatively high LOD of approximately 0.1 mg $FL^{-1}$ for the HPLC–ICP-MS/MS a large F-peak is shown in FP 30, which indicates the ppm level of concentration, but which could not be identified by ESI-MS. This illustrates the limitations of depending on ESI-MS methods for non-target analysis of PFAS. Using a hard ionisation source and fluorine-specific detection such as ICP-MS/MS is advantageous despite their inability to identify the molecular formulae of the organofluorine compound. This means that the ICP-MS/MS detection illustrates the occurrence of a discrete low molecular mass organofluorine which cannot be identified by molecular mass spectrometry (i.e. the unidentified PFAS in the EOF of FP 30 were not only extractable polymers and fluorine-containing nano- and microparticles).

The rest of the samples did not show any peaks at $m/z$ 157 ($BaF^+$) due to the low sensitivity for the fluorine-specific detection, illustrating the limitation of the current ICP-MS technology for fluorine detection.

Comparing LC–HRMS to LC–MS/MS, more PFAS were detected using LC–MS/MS, since it is more sensitive (Table S7).

The sum of the identified PFAS (∑PFAS) using LC–MS/MS was always less than 0.5% when compared to the EOF for the KSA and UK samples (Figure S8). Other PFAS not on the target list such as fluorotelomer polyfluoroalkyl phosphate esters (PAP) might make up some of the EOF, since this seems to be common in high concentration in FCM (Dueñas-Mas et al. 2023), although our non-target analysis approach with HRMS and ICP-MS/MS did not detect them. Furthermore, as mentioned in previous studies, the low % identification of ∑PFAS shows the limitations of using the electrospray ionisation and the target analysis approach, since it does not capture the neutral PFAS intentionally added to the FP material (Schultes et al. 2019). Most of the PFAS would be in polymeric form and might not be completely extractable into the EOF. However,

**Table 2.** Accurate masses recorded by ESI-HRMS which formed peaks at the same retention time as fluorine in ICP-MS/MS with the proposed molecular formulas and associated errors including in-source fragmentation.

| Peak no. | Expt $[M-H]^-$ | Ion formula | Found $[M-H]^-$ | Error (ppm) | Rdb | RT (min) |
|---|---|---|---|---|---|---|
| 1a | 498.304 | $C_8F_{17}O_3S$ | 498.9302 | −0.40 | 2 | 11.1 |
| | 79.9574 | $O_3S$ | 79.9574 | 0.71 | | |
| | 98.9559 | $O_3FS$ | 98.9558 | 1.7 | | |
| 2a–c | 525.9781 | $C_{10}H_5F_{17}NO_2S$ | 525.9775 | −1.2 | 2 | 18.8 |
| 1b | 498.9311 | $C_8F_{17}O_3S$ | 498.9302 | −1.8 | 2 | 11.3 |
| | 79.9577 | $O_3S$ | 79.9574 | 3.7 | | |

Rdb: Rings and Double Bond Equivalents..

when they are extracted then they may not be ionisable or do not elute from the HPLC-column. Another explanation could be the presence of unreacted monomers and degradation products from polymers [e.g. fluoro-telomere olefin, fluorotelomers (FTOH)], which are extractable but also not detected by non-targeted and targeted analysis using ESI-MS.

In all of the regulatory actions, only certain PFAS have been addressed, mostly PFCA (perfluoroalkyl carboxylic acids) and PFSA (perfluoroalkyl sulphonates). However, as shown in this study and Schultes et al. (2019), there is a large quantity of the EOF which is not explainable with PFCA and PFSA detection.

Although currently there is still not sufficient evidence for most PFAS on their adverse effect on human health and the environment, there is an indication that the entire PFAS family could have similar consequences (ECHA 2023). Therefore, a broader chemical scope that would encompass production, placing on the market, and usage is necessary. In March 2023, ECHA published Annex XV Restriction Report for PFAS (ECHA 2023) which would encompass a broader chemical scope of PFAS, hence when the detection limit for fluorine detection can be improved ICP-MS/MS would be the method of choice of analysing PFAS to identify non-polymeric PFAS eluting from the HPLC column as shown here. The restriction also specifies maximum allowed quantities in articles placed on the market: 25 $\mu g\,kg^{-1}$ for individual PFAS and 250 $\mu g\,kg^{-1}$ for the sum of PFAS measured with targeted analysis, and 50 $mg\,kg^{-1}$ if the targeted analysis is not applicable, in the case of fluoropolymers. If TF content in fluoropolymers exceeds 50 mg $F\,kg^{-1}$, proof of whether or not the measured fluorine is part of PFAS should also be provided to authorities.

## Conclusions

It has been shown that FCM from different countries contain a diverse PFAS spectrum but with similar quantity. FCM bought in Sweden before 2018 show a higher diversity and more C8 representatives among the PFAS compounds than those bought after 2018 in the UK and in KSA. Whether this is due to the adding of PFOA to the banned PFAS in the Stockholm POP list or a geographical difference is unclear. FCM from the later countries were subject to total EOFs and it was obvious that the target PFAS account for only traces of organofluorines in the EOF. Non-target approaches using LC coupled to HRMS and to ICP-MS could reveal only some unidentified PFAS, but they failed to explain the identity of the majority of organofluorines in the EOF. Hence, it is still important to develop new analytical methods which could target all PFAS regardless of their properties (polymeric or particulates or volatile and neutral or ionic). This is very important in the context of newly proposed legislation (e.g. the ECHA proposal from 2023) which tries to phase out all PFAS to which humans can be exposed.

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